

Solution

- The total-time minimization (with $a_1=1, a_2>0$) leads to a corner solution. In fact one checks that $a_2>1$ (the series defining a_2 converges to $1+O(72^{-10^{24}})$, since $\sum_{k=3}^{\infty} 2^{-k}=1/4$ ¹). Hence the fastest way to achieve dosage 1 is to use only the second (faster) therapy. The unique minimiser is $(x_1, x_2)=(0, 1/a_2)$.
- We compute the infinity-norm distance from $(0.1, 0.9)$ to the minimiser $(0, 1/a_2)$: $\|(0.1, 0.9)-(0, 1/a_2)\|_{\infty} = \max(|0.1-0|, |0.9-1/a_2|)$ (by definition $\|v\|_{\infty} = \max_i |v_i|$ ²). Since $a_2 \approx 1 + \epsilon$, one has $1/a_2 \approx 1$, so $|0.9-1/a_2| \approx 0.1$. Thus $\|(0.1, 0.9)-(0, 1/a_2)\|_{\infty} \approx 0.10$.

Because this distance is about 0.10 , it is indeed ≤ 0.15 . Therefore **yes** – the point $(0.1, 0.9)$ lies at most 0.15 (in fact about 0.10) away from the true minimiser.

Answer: Yes, since $\|(0.1, 0.9)-(0, 1/a_2)\|_{\infty} \approx 0.10 < 0.15$ ¹ ².

¹ Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series

² Norm (mathematics) - Wikipedia

[https://en.wikipedia.org/wiki/Norm_\(mathematics\)](https://en.wikipedia.org/wiki/Norm_(mathematics))